

Types of Machine Learning, Supervised learning: Classification, Regression, Unsupervised learning, Generative and Discriminative Models, Some basic concepts in machine learning, The Machine Learning Process, Reinforcement Learning.

Definition of machine learning

Arthur Samuel, an early American leader in the field of computer gaming and artificial intelligence, **coined the term “Machine Learning” in 1959** while at IBM. He defined machine learning as **“the field of study that gives computers the ability to learn without being explicitly programmed.”** However, there is no universally accepted definition for machine learning. Different authors define the term differently.

Two more definitions.

1. Machine learning is programming computers to optimize a performance criterion using example data or past experience. We have a model defined up to some parameters, and learning is the execution of a computer program to optimize the parameters of the model using the training data or past experience. The model may be predictive to make predictions in the future, or descriptive to gain knowledge from data, or both
2. The field of study known as machine learning is concerned with the question of how to construct computer programs that automatically improve with experience

Definition of learning

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks T , as measured by P , improves with experience E .

Examples

i) Handwriting recognition learning problem

- Task T : Recognizing and classifying handwritten words within images
- Performance P : Percent of words correctly classified
- Training experience E : A dataset of handwritten words with given classifications

ii) A robot driving learning problem

- Task T : Driving on highways using vision sensors
- Performance measure P : Average distance traveled before an error
- Training experience: A sequence of images and steering commands recorded while observing a human driver

iii) A chess learning problem

- Task T : Playing chess
- Performance measure P : Percent of games won against opponents

- Training experience E: Playing practice games against itself

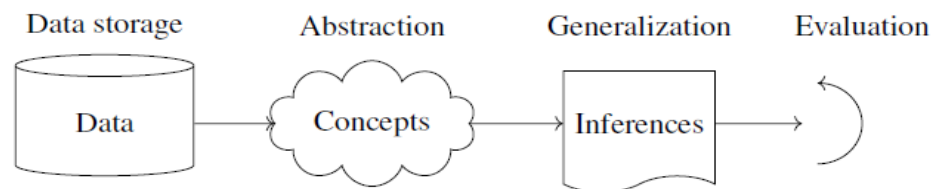
A computer program which learns from experience is called a machine learning program or simply a learning program. Such a program is sometimes also referred to as a learner.



How machines learn?

Basic components of learning process

The learning process, whether by a human or a machine, can be divided into four components, namely, data storage, abstraction, generalization and evaluation. Figure illustrates the various components and the steps involved in the learning process.



1. Data storage

Facilities for storing and retrieving huge amounts of data are an important component of the learning process. Humans and computers alike utilize data storage as a foundation for advanced reasoning.

- In a human being, the data is stored in the brain and data is retrieved using electrochemical signals.
- Computers use hard disk drives, flash memory, random access memory and similar devices to store data and use cables and other technology to retrieve data.

2. Abstraction

The second component of the learning process is known as abstraction.

Abstraction is the process of extracting knowledge about stored data. This involves creating general concepts about the data as a whole. The creation of knowledge involves application of known models and creation of new models.

The process of fitting a model to a dataset is known as training. When the model has been trained, the data is transformed into an abstract form that summarizes the original information.

3. Generalization

The third component of the learning process is known as generalisation.

The term generalization describes the process of turning the knowledge about stored data into a form that can be utilized for future action. These actions are to be carried out on tasks that are similar, but

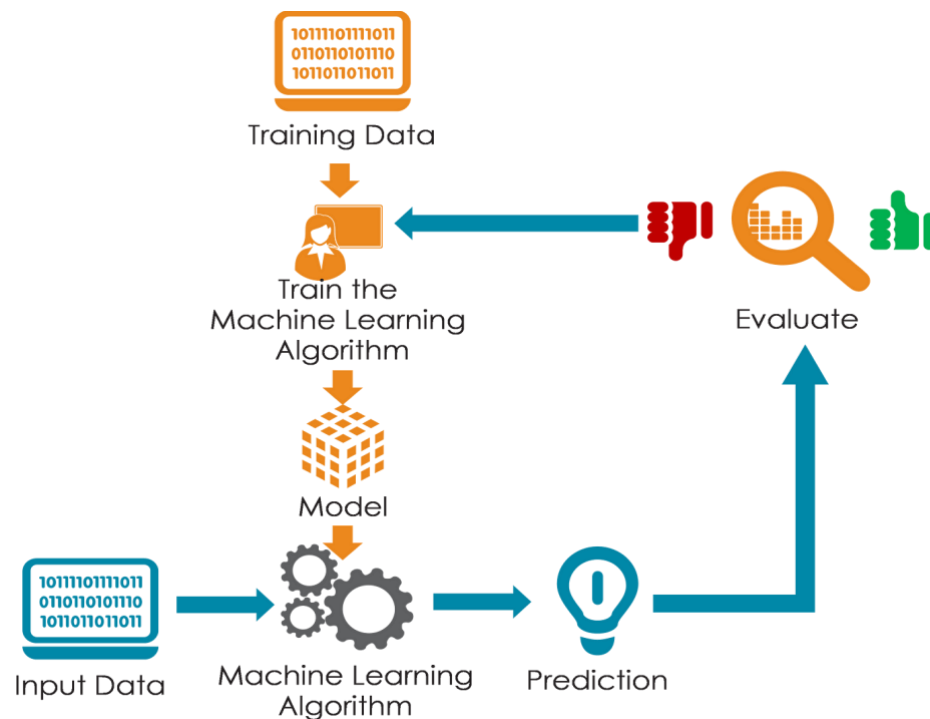
not identical, to those what have been seen before. In generalization, the goal is to discover those properties of the data that will be most relevant to future tasks.

4. Evaluation

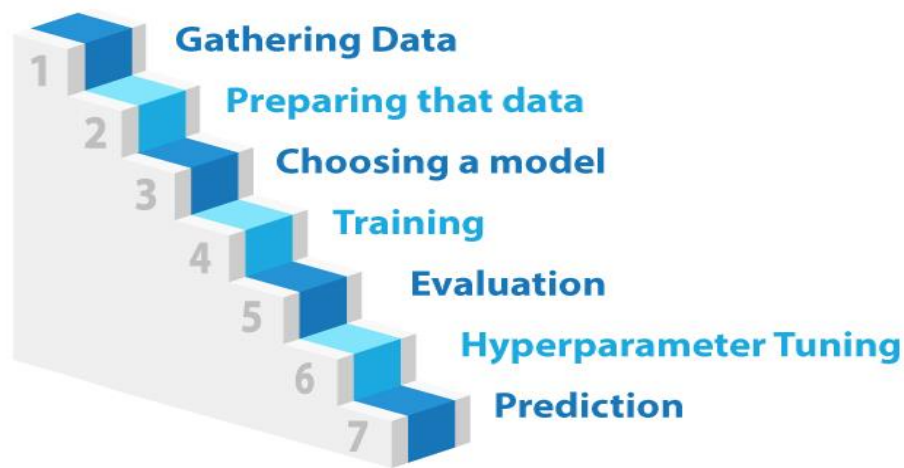
Evaluation is the last component of the learning process.

It is the process of giving feedback to the user to measure the utility of the learned knowledge.

This feedback is then utilised to effect improvements in the whole learning process.

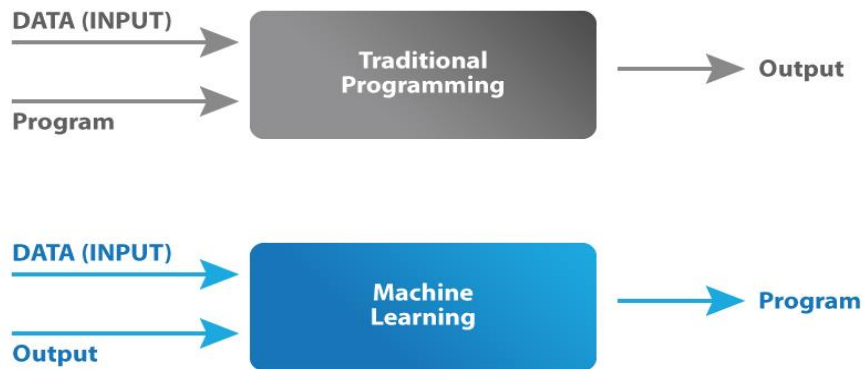


7 steps of Machine Learning



Difference between Traditional and Machine Learning Programming

- In **traditional programming**, we could feed the input data and a well written and tested program into a machine to generate the output
- In **machine learning** input data along with the output is fed into the machine during the learning phase and it works out a program for itself.



Understanding data

The different types and forms of data that are encountered in the machine learning process are.

Unit of observation

By a unit of observation we mean the smallest entity with measured properties of interest for a study.

Examples

- A person, an object or a thing
- A time point
- A geographic region
- A measurement

Sometimes, units of observation are combined to form units such as person-years

Examples and features

Datasets that store the units of observation and their properties can be imagined as collections of data consisting of the following:

• Examples

- ✓ An “example” is an instance of the unit of observation for which properties have been recorded.
- ✓ An “example” is also referred to as an “instance”, or “case” or “record.” (It may be noted that the word “example” has been used here in a technical sense.)

• Features

A “feature” is a recorded property or a characteristic of examples. It is also referred to as “attribute”, or “variable” or “feature.”

Examples for “examples” and “features”

1. Cancer detection

Consider the problem of developing an algorithm for detecting cancer. In this study we note the following.

- (a) The units of observation are the patients.
- (b) The examples are members of a sample of cancer patients.
- (c) The following attributes of the patients may be chosen as the features:
 - ✓ gender
 - ✓ age
 - ✓ blood pressure
 - ✓ the findings of the pathology report after a biopsy

2. Pet selection

Suppose we want to predict the type of pet a person will choose.

- (a) The units are the persons.
- (b) The examples are members of a sample of persons who own pets.
- (c) The features might include age, home region, family income, etc. of persons who own pets.

year	model	price	mileage	color	transmission
2011	SEL	21992	7413	Yellow	AUTO
2011	SEL	20995	10926	Gray	AUTO
2011	SEL	19995	7351	Silver	AUTO
2011	SEL	17809	11613	Gray	AUTO
2012	SE	17500	8367	White	MANUAL
2010	SEL	17495	25125	Silver	AUTO
2011	SEL	17000	27393	Blue	AUTO
2010	SEL	16995	21026	Silver	AUTO
2011	SES	16995	32655	Silver	AUTO

Figure Example for “examples” and “features” collected in a matrix format (data relates to automobiles and their features)

Spam e-mail

Let it be required to build a learning algorithm to identify spam e-mail.

- (a) The unit of observation could be an e-mail messages.
- (b) The examples would be specific messages.
- (c) The features might consist of the words used in the messages.

Examples and features are generally collected in a “matrix format”.

Different forms of data

1. Numeric data

If a feature represents a characteristic measured in numbers, it is called a numeric feature.

2. Categorical or nominal

A categorical feature is an attribute that can take on one of a limited, and usually fixed, number of possible values on the basis of some qualitative property. A categorical feature is also called a nominal feature.

3. Ordinal data

This denotes a nominal variable with categories falling in an ordered list. Examples include clothing sizes such as small, medium, and large, or a measurement of customer satisfaction on a scale from “not at all happy” to “very happy.”

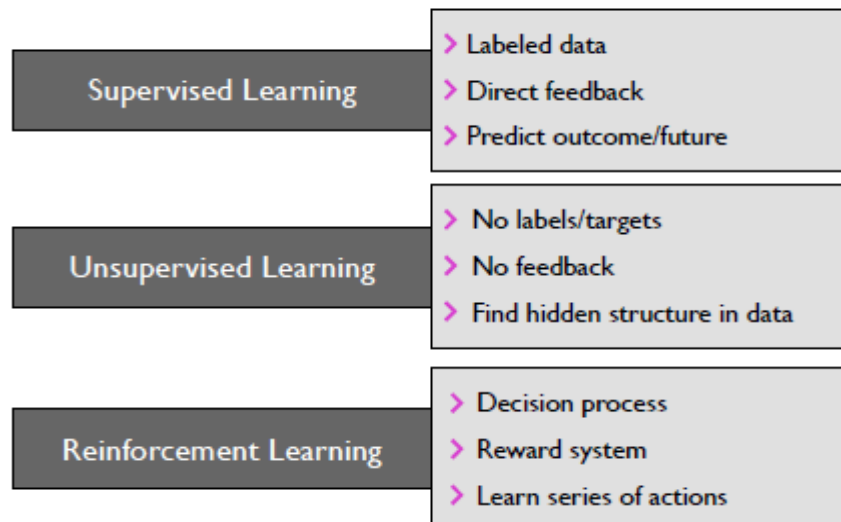
Examples

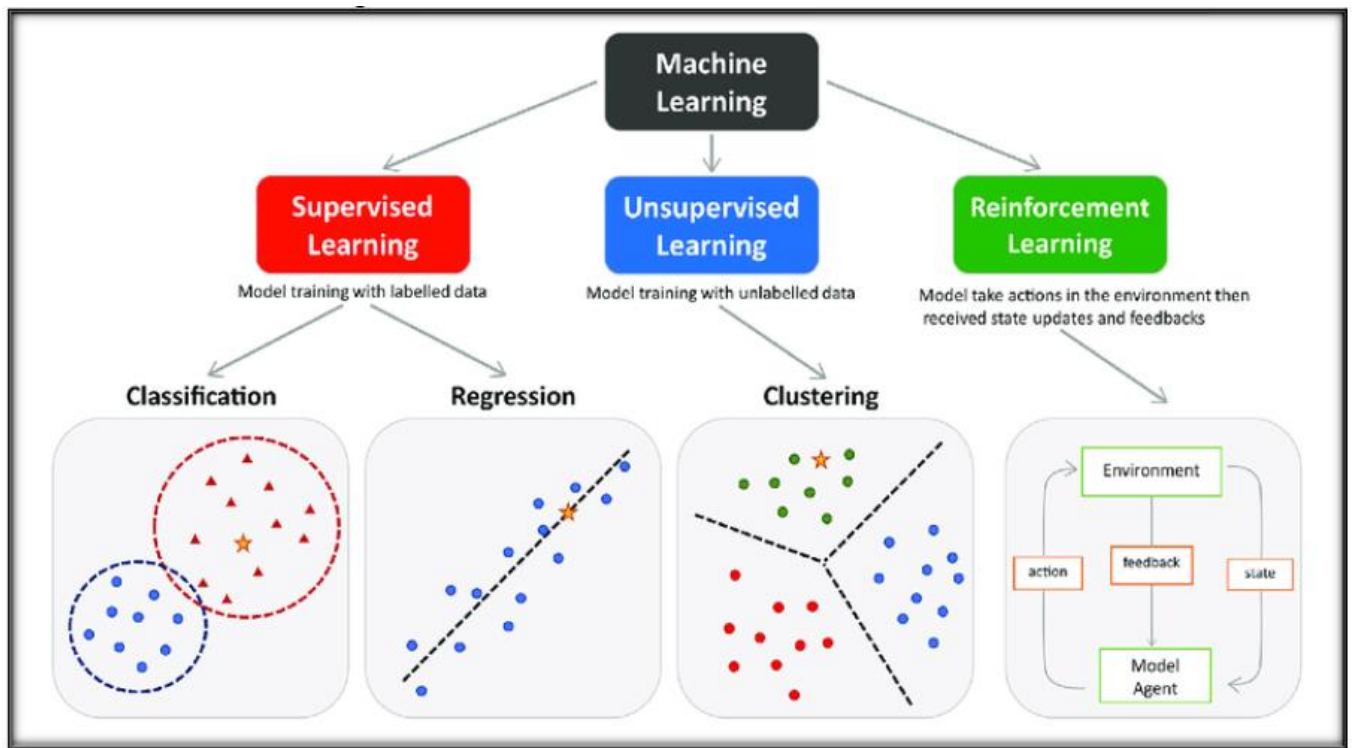
In the data given the features “year”, “price” and “mileage” are numeric and the features “model”, “color” and “transmission” are categorical.

1.1 TYPES OF MACHINE LEARNING

Based on the methods and way of learning, machine learning is divided into mainly four types, which are:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Semi-supervised learning
4. Reinforcement Learning





1. Supervised learning

Supervised learning is the subcategory of machine learning that focuses on learning a **classification or regression model**, that is, learning from labeled training data (i.e., inputs that also contain the desired outputs or targets; basically, "examples" of what we want to predict).

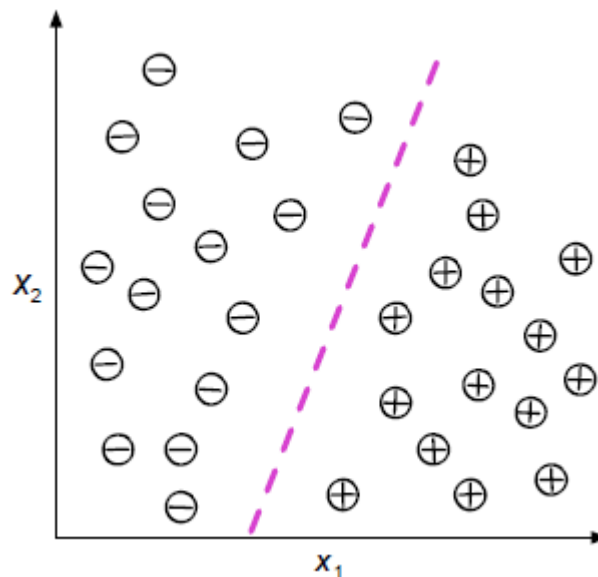


Figure: Illustration of a binary classification problem (plus, minus) and two feature variable (x_1 and x_2). :

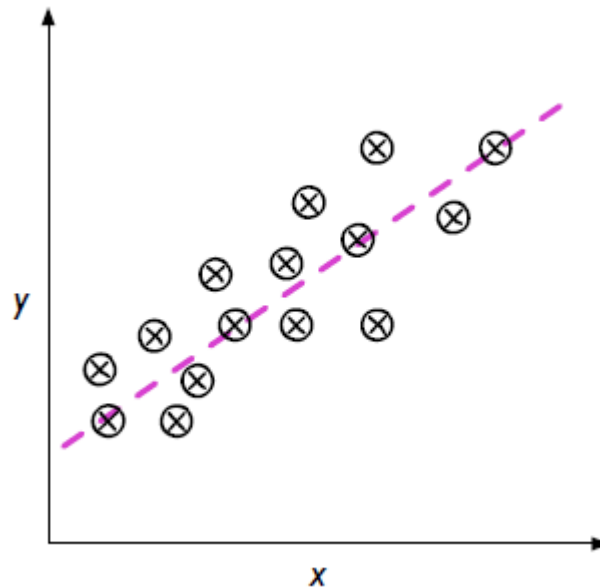


Figure: Illustration of a linear regression model with one feature variable (x_1) and the target variable y . The dashed-line indicates the functional form of the linear regression model.

2. Unsupervised learning

In contrast to supervised learning, unsupervised learning is a branch of machine learning that is concerned with unlabeled data. Common tasks in unsupervised learning are clustering analysis (assigning group memberships) and dimensionality reduction (compressing data onto a lower-dimensional subspace or manifold).

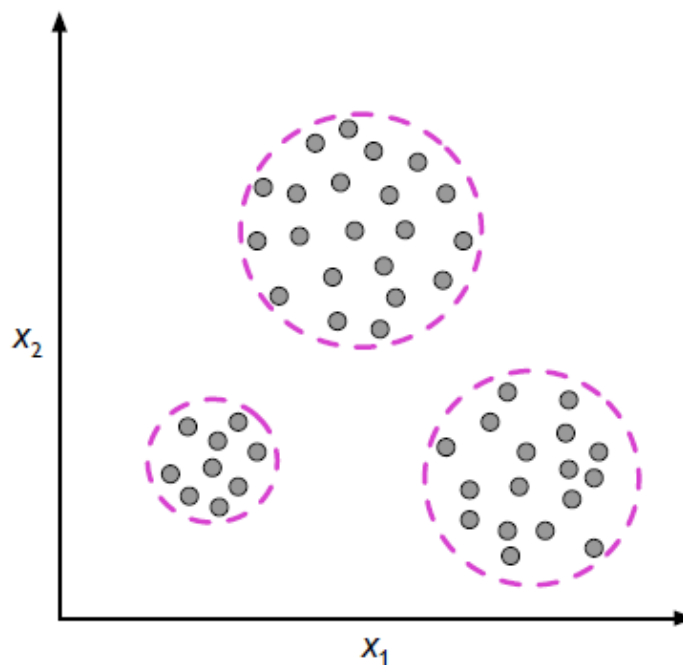


Figure Illustration of clustering, where the dashed lines indicate potential group membership assignments of unlabeled data points.

3. Semi-Supervised Learning

Loosely speaking, semi-supervised learning can be described as a mix between supervised and unsupervised learning. In semi-supervised learning tasks, some training examples contain outputs, but some do not. We then use the labeled training subset to label the unlabeled portion of the training set, which we then also utilize for model training.

- Semi-Supervised learning is a type of Machine Learning algorithm that lies between Supervised and Unsupervised machine learning.
- It represents the intermediate ground between Supervised (With Labelled training data) and Unsupervised learning (with no labelled training data) algorithms and uses the combination of labelled and unlabeled datasets during the training period.
- Although Semi-supervised learning is the middle ground between supervised and unsupervised learning and operates on the data that consists of a few labels, it mostly consists of unlabeled data.
- As labels are costly, but for corporate purposes, they may have few labels.
- It is completely different from supervised and unsupervised learning as they are based on the presence & absence of labels.

To overcome the drawbacks of supervised learning and unsupervised learning algorithms, the concept of Semi-supervised learning is introduced.

- The main aim of semi-supervised learning is to effectively use all the available data, rather than only labelled data like in supervised learning.
- Initially, similar data is clustered along with an unsupervised learning algorithm, and further, it helps to label the unlabeled data into labelled data.
- It is because labelled data is a comparatively more expensive acquisition than unlabeled data.
- We can imagine these algorithms with an example.
- Supervised learning is where a student is under the supervision of an instructor at home and college. Further, if that student is self-analysing the same concept without any help from the instructor, it comes under unsupervised learning.
- Under semi-supervised learning, the student has to revise himself after analyzing the same concept under the guidance of an instructor at college.

4. Reinforcement learning

Reinforcement is the process of learning from rewards while performing a series of actions. In reinforcement learning, we do not tell the learner or agent, for example, a (ro)bot, which action to take but merely assign a reward to each action and/or the overall outcome. Instead of having "correct/false" label for each step, the learner must discover or learn a behavior that maximizes the reward for a series

of actions. In that sense, it is not a supervised setting and somewhat related to unsupervised learning; however, reinforcement learning really is its own category of machine learning. Reinforcement learning will not be covered further in this class.

Typical applications of reinforcement learning involve playing games (chess, Go, Atari video games) and some form of robots, e.g., drones, warehouse robots, and more recently selfdriving cars.

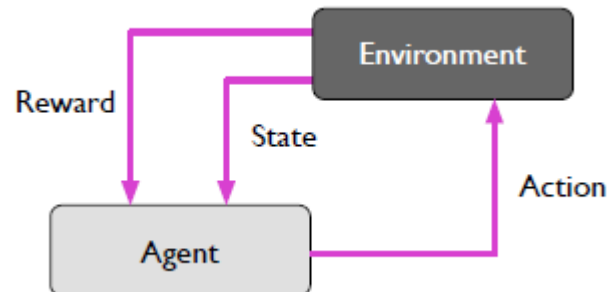


Figure Illustration of reinforcement learning

1.2 SUPERVISED LEARNING

Supervised learning

- Supervised learning is a type of machine learning that uses labeled data to train machine learning models.

Working:

- Supervised learning algorithms takes labeled inputs and map them to the known outputs, which means you already know the target variable.
- Supervised Learning methods need external supervision to train machine learning models. Hence, the name was supervised. They need guidance and additional information to return the desired result.
- In supervised learning, models are trained using labelled dataset, where the model learns about each type of data. Once the training process is completed, the model is tested on the basis of test data (a subset of the training set), and then it predicts the output.

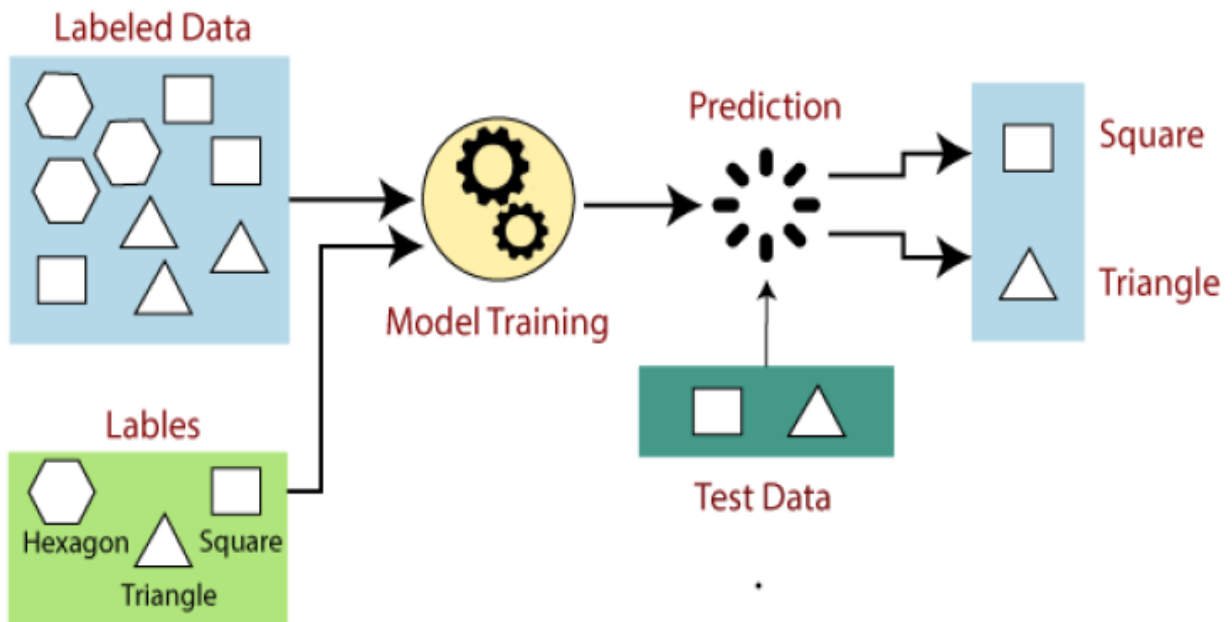


Figure : Supervised learning

Suppose we have a dataset of different types of shapes which includes square, rectangle, triangle, and Polygon. Now the first step is that we need to train the model for each shape.

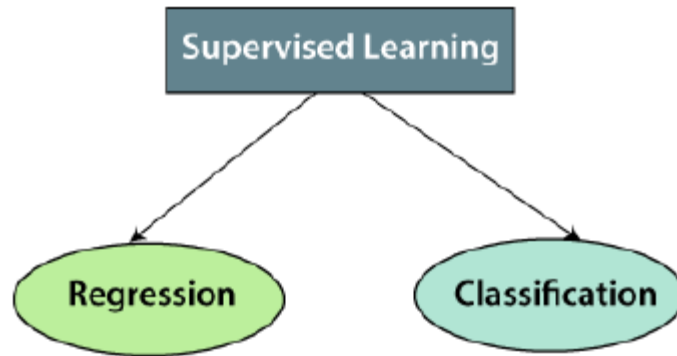
- If the given shape has four sides, and all the sides are equal, then it will be labelled as a Square.
- If the given shape has three sides, then it will be labelled as a triangle.
- If the given shape has six equal sides then it will be labelled as hexagon. Now, after training, we test our model using the test set, and the task of the model is to identify the shape. The machine is already trained on all types of shapes, and when it finds a new shape, it classifies the shape on the bases of a number of sides, and predicts the output.

Steps Involved in Supervised Learning:

- First Determine the type of training dataset
- Collect/Gather the labelled training data.
- Split the training dataset into training dataset, test dataset, and validation dataset.
- Determine the input features of the training dataset, which should have enough knowledge so that the model can accurately predict the output.
- Determine the suitable algorithm for the model, such as support vector machine, decision tree, etc.
- Execute the algorithm on the training dataset. Sometimes we need validation sets as the control parameters, which are the subset of training datasets.
- Evaluate the accuracy of the model by providing the test set. If the model predicts the correct output, which means our model is accurate.

Types of supervised Machine learning Algorithms:

Supervised learning can be further divided into two types of problems:



a. Regression: -

Regression algorithms are used if there is a relationship between the input variable and the output variable. It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc.

Popular Regression algorithms which come under supervised learning:

- Linear Regression
- Regression Trees
- Non-Linear Regression
- Bayesian Linear Regression
- Polynomial Regression

b. Classification: -

Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.

Popular classification algorithms which come under supervised learning:

- Random Forest
- Decision Trees
- Logistic Regression
- Support vector Machines

Advantages of Supervised Learning: -

- With the help of supervised learning, the model can predict the output on the basis of prior experiences.
- In supervised learning, we can have an exact idea about the classes of objects.
- Supervised learning model helps us to solve various real-world problems such as fraud detection, spam filtering, etc.

Disadvantages of supervised Learning:

- Supervised learning models are not suitable for handling the complex tasks.
- Supervised learning cannot predict the correct output if the test data is different from the training dataset.
- Training required lots of computation times.
- In supervised learning, we need enough knowledge about the classes of object.

Algorithms:

Some of the most popularly used supervised learning algorithms are:

- Linear Regression
- Logistic Regression
- Support Vector Machine
- K Nearest Neighbor
- Decision Tree
- Random Forest
- Naive Bayes
- Neural Networks

Some applications of Supervised Learning include:

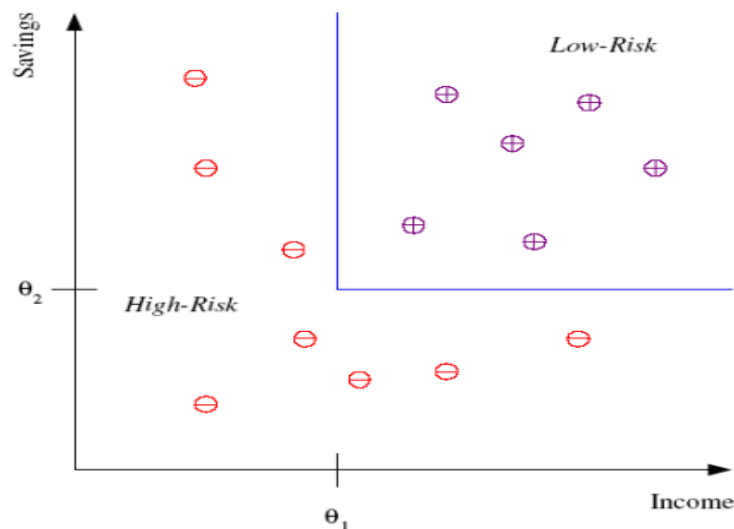
1. It classifies email spam Detection by teaching a model of what mail is spam and not spam.
2. Speech recognition where you teach a machine to recognize your voice.
3. Object Recognition by showing a machine what an object looks like and having it pick that object from among other objects.

1.3 CLASSIFICATION

The classification problem consists of taking input vectors and deciding which of N classes they belong to, based on training from exemplars of each class

Example1: Credit scoring Differentiating between low-risk and high-risk customers from their income and savings

In credit scoring, the bank calculates the risk given the amount of credit and the information about the customer. The information about the customer includes data we have access to and is relevant in calculating his or her financial capacity—namely, income, savings, collaterals, profession, age, past financial history, and so forth. The bank has a record of past loans containing such customer data and whether the loan was paid back or not. From this data of particular applications, the aim is to infer a general rule coding the association between a customer's attributes and his risk. That is, the machine learning system fits a model to the past data to be able to calculate the risk for a new application and then decides to accept or refuse it accordingly.



The most important point about the classification problem is that it is **discrete**—each example belongs to precisely one class, and the set of classes covers the whole possible output space.

This is an example of a classification problem where there are two classes: **low-risk and high-risk customers**. The information about a customer makes up the input to the classifier whose task is to assign the input to one of the two classes. After training with the past data, a classification rule learned may be of the form

IF income > θ_1 AND savings > θ_2 THEN low-risk ELSE high-risk

Discriminant is a function that separates the examples of different classes.

Example 2: Coin classification

We train the classifier to recognise all **New Zealand coins**, but what if a **British coin** is put into the machine? In that case, the classifier will identify it as the New Zealand coin that is closest to it in appearance, but this is not really what is wanted: rather, the classifier should identify that it is not one of the coins it was trained on. This is called novelty detection. For now we'll assume that we will not receive inputs that we cannot classify accurately.

Let's consider **how to set up a coin classifier**. When the coin is pushed into the slot, the machine takes a few measurements of it. These could include the diameter, the weight, and possibly the shape, and are the features that will generate our input vector. In this case, our input vector will have three elements, each of which will be a number showing the measurement of that feature (choosing a number to represent the shape would involve an encoding, for example that 1=circle, 2=hexagon, etc.). Of course, there are many other features that we could measure. If our vending machine included an atomic absorption spectroscope, then we could estimate the density of the material and its composition, or if it had a camera, we could take a photograph of the coin and feed that image into the classifier.

The question of which features to choose is not always an easy one. We don't want to use too many inputs, because that will make the training of the classifier take longer (and also, as the number of input dimensions grows, the number of datapoints required increases faster; this is known as the curse of dimensionality, but we need to make sure that we can reliably separate the classes based on those features.



FIGURE The New Zealand coins

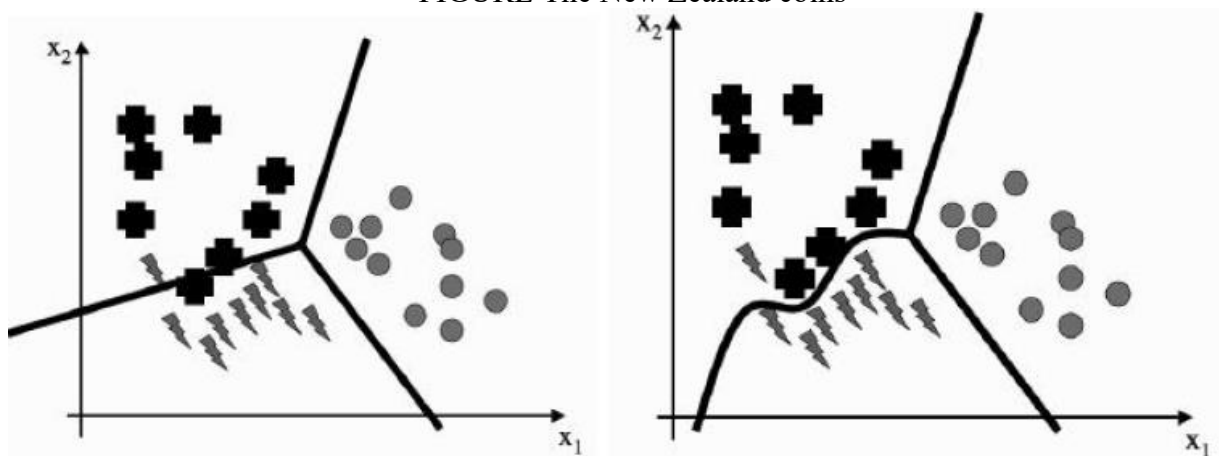


FIGURE *Left*: A set of straight line decision boundaries for a classification problem. *Right*: An alternative set of decision boundaries that separate the pluses from the lightening strikes better, but requires a line that isn't straight..

For example, if we tried to separate coins based only on colour, we wouldn't get very far, because the 20 ¢ and 50 ¢ coins are both silver and the \$1 and \$2 coins both bronze. However, if we use colour and diameter, we can do a pretty good job of the coin classification problem for NZ coins. There are some features that are entirely useless. For example, knowing that the coin is circular doesn't tell us anything about NZ coins, which are all circular. In other countries, though, it could be very useful. The methods of performing classification that we will see during this book are very different in the ways that they learn about the solution; in essence they aim to do the same thing: find decision boundaries that can be used to separate out the different classes. Given the features that are used as inputs to the classifier, we need to identify some values of those features that will enable us to decide which class the current input is in. Figure shows a set of 2D inputs with three different classes shown,

and two different decision boundaries; on the left they are straight lines, and are therefore simple, but don't categorise as well as the non-linear curve on the right. Now that we have seen these two types of problem, let's take a look at the whole process of machine learning from the practitioner's viewpoint.

Classification: Applications

- **Face recognition:** Pose, lighting, occlusion (glasses, beard), make-up, hair style
- **Character recognition:** Different handwriting styles.
- **Speech recognition:** Temporal dependency.
- **Medical diagnosis:** From symptoms to illnesses
- **Biometrics:** Recognition/authentication using physical and/or behavioral characteristics:
Face, iris, signature, etc

1.4 REGRESSION

- In machine learning, a regression problem is the problem of predicting the value of a numeric variable based on observed values of the variable.
- The value of the output variable may be a number, such as an integer or a floating point value. These are often quantities, such as amounts and sizes.
- The input variables may be discrete or real-valued.

Example Consider the data on car prices

Price (US\$)	Age (years)	Distance (KM)	Weight (pounds)
13500	23	46986	1165
13750	23	72937	1165
13950	24	41711	1165
14950	26	48000	1165
13750	30	38500	1170
12950	32	61000	1170
16900	27	94612	1245
18600	30	75889	1245
21500	27	19700	1185
12950	23	71138	1105

Table Prices of used cars: example data for regression

Suppose we are required to estimate the price of a car aged 25 years with distance 53240 KM and weight 1200 pounds. This is an example of a regression problem because we have to predict the value of the numeric variable "Price".

General approach

Let x denote the set of input variables and y the output variable. In machine learning, the general approach to regression is to assume a model, that is, some mathematical relation between x and y , involving some parameters say in the following form:

$$y = f(x, \theta)$$

The function $f(x, \theta)$ is called the regression function. The machine learning algorithm optimizes the parameters in the set such that the approximation error is minimized; that is, the estimates of the values of the dependent variable y are as close as possible to the correct values given in the training set.

Regression: Applications

- Loan Default Prediction
- House Price Prediction
- Stock Market Prediction
- Market Sales Forecasting
- Advertising

1.5 UNSUPERVISED LEARNING

Unsupervised learning

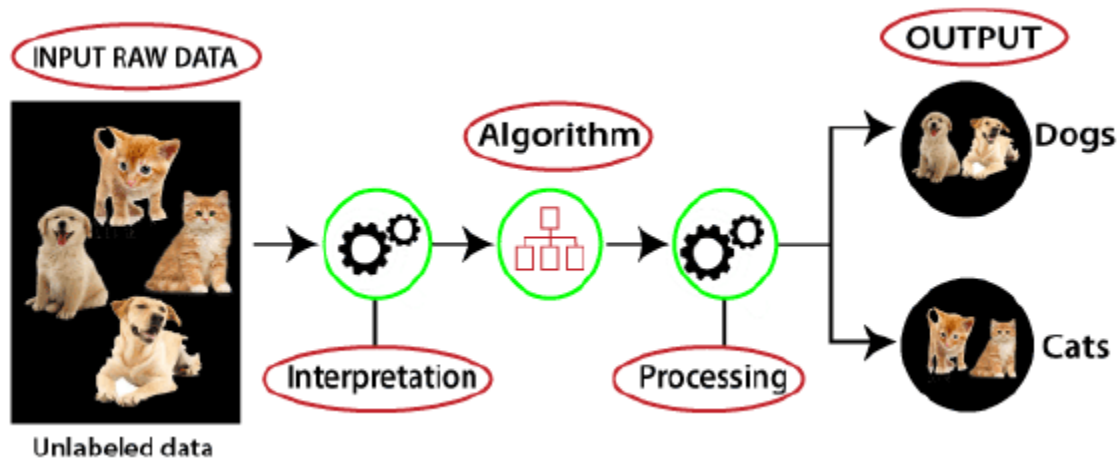
Unsupervised learning is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision.

The goal of unsupervised learning is to find the underlying structure of dataset, group that data according to similarities, and represent that dataset in a compressed format.

Working of Unsupervised Learning

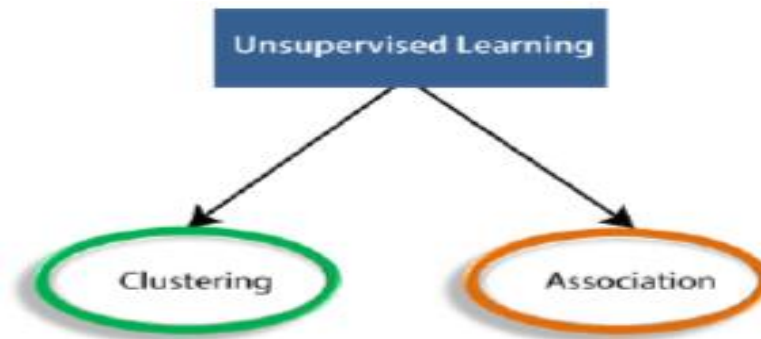
Working of unsupervised learning can be understood by the below diagram:

Here, we have taken an unlabeled input data, which means it is not categorized and corresponding outputs are also not given. Now, this unlabeled input data is fed to the machine learning model in order to train it. Firstly, it will interpret the raw data to find the hidden patterns from the data and then will apply suitable algorithms such as k-means clustering etc. Once it applies the suitable algorithm, the algorithm divides the data objects into groups according to the similarities and difference between the objects.



Types of Unsupervised Learning Algorithm:

The unsupervised learning algorithm can be further categorized into two types of problems:



1. Clustering: Clustering is a method of grouping the objects into clusters such that objects with most similarities remain in a group and have less or no similarities with the objects of another group. Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.

2. Association: - An association rule is an unsupervised learning method which is used for finding the relationships between variables in the large database. It determines the set of items that occurs together in the dataset. Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item. A typical example of Association rule is Market Basket Analysis.

Unsupervised Learning algorithms:

- K-means clustering
- Hierarchical clustering
- Anomaly detection
- Principle Component Analysis
- Independent Component Analysis
- Apriori algorithm
- Singular value decomposition

Advantages of Unsupervised Learning: -

- Unsupervised learning is used for more complex tasks as compared to supervised learning because, in unsupervised learning, we don't have labeled input data.
- Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.

Disadvantages of Unsupervised Learning: -

- Unsupervised learning is intrinsically more difficult than supervised learning as it does not have corresponding output.
- The result of the unsupervised learning algorithm might be less accurate as input data is not labeled, and algorithms do not know the exact output in advance.

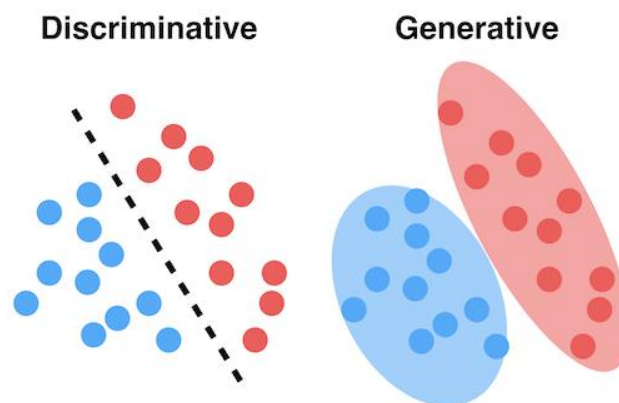
Supervised learning Vs Unsupervised learning

In supervised learning, the aim is to learn a mapping from the input to an output whose correct values are provided by a supervisor. In unsupervised learning, there is no such supervisor and we only have input data.

1.6 GENERATIVE AND DISCRIMINATIVE MODELS

Machine learning models can be classified into two types: **Discriminative and Generative**.

- In simple words, a **discriminative model** makes predictions on unseen data based on conditional probability and can be used either for classification or regression problem statements.
- On the contrary, a **generative model** focuses on the distribution of a dataset to return a probability for a given example.



They are related to known effects of causal direction, classification vs. inference learning, and observational vs. feedback learning.

Problem Formulation

Suppose we are working on a classification problem where our task is to decide if an email is spam or not spam based on the words present in a particular email. To solve this problem, we have a joint model over.

- Labels: $Y=y$, and
- Features: $X=\{x_1, x_2, \dots, x_n\}$

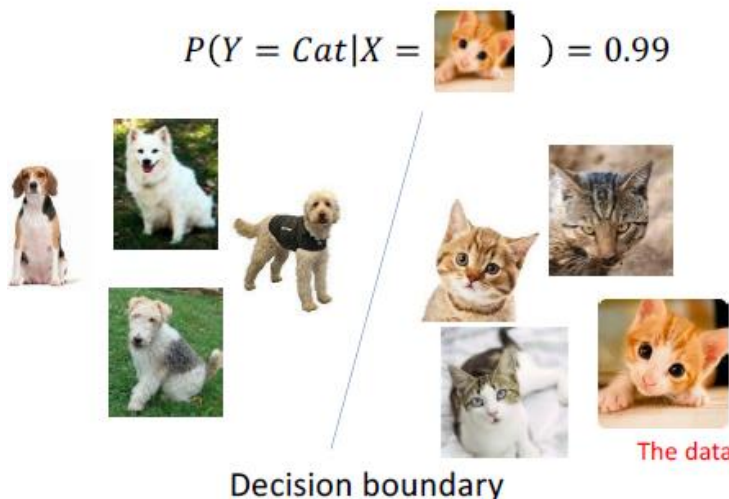
Therefore, the joint distribution of the model can be represented as

$$p(Y,X) = P(y, x_1, x_2, \dots, x_n)$$

Now, our goal is to estimate the probability of spam email i.e., $P(Y=1|X)$. Both generative and discriminative models can solve this problem but in different ways.

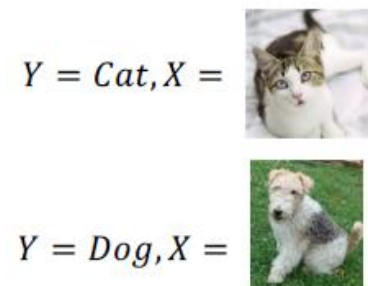
Discriminative models: classify data

finding **conditional distribution** $P(Y|X)$



Generative models: generate data

finding **joint distribution** $P(Y, X)$



- Discriminative models do not model/learn the probability distribution of data $p(x)$ and find the decision boundary directly to form $p(y|x)$
- Generative models need to
 - first model/learn the probability distribution of data $p(x)$
 - and the joint probability distribution $p(x, y)$
 - and the estimated the conditional probability $p(y|x) = \frac{p(x,y)}{p(x)}$

1.6.1 The Approach of Discriminative Models

In the case of discriminative models, to find the probability, they directly assume some functional form for $P(Y|X)$ and then estimate the parameters of $P(Y|X)$ with the help of the training data.

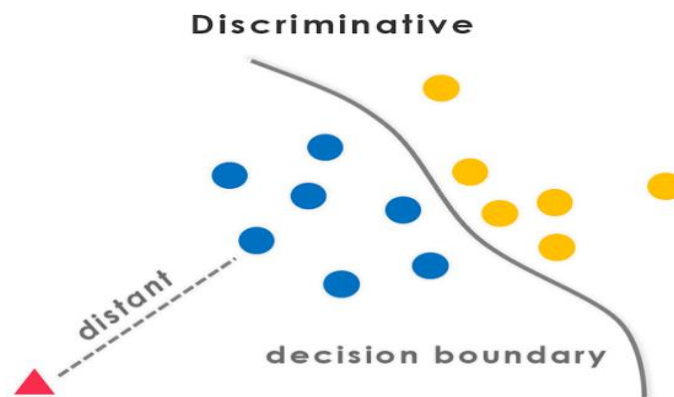
Discriminative Models

The discriminative model refers to a class of models used in Statistical Classification, mainly used for supervised machine learning. These types of models are also known as conditional models since they learn the boundaries between classes or labels in a dataset.

Discriminative models focus on modeling the decision boundary between classes in a classification problem. The goal is to learn a function that maps inputs to binary outputs, indicating the class label of the input. Maximum likelihood estimation is often used to estimate the parameters of the discriminative model, such as the coefficients of a logistic regression model or the weights of a neural network.

Discriminative models (just as in the literal meaning) separate classes instead of modeling the conditional probability and don't make any assumptions about the data points. But these models are not capable of generating new data points. Therefore, the ultimate objective of discriminative models is to separate one class from another.

If we have some outliers present in the dataset, discriminative models work better compared to generative models i.e., discriminative models are more robust to outliers. However, one major drawback of these models is the misclassification problem, i.e., wrongly classifying a data point.



The Mathematics of Discriminative Models

Training discriminative classifiers or discriminant analysis involves estimating a function $f: X \rightarrow Y$, or probability $P(Y|X)$

- Assume some functional form for the probability, such as $P(Y|X)$
- With the help of training data, we estimate the parameters of $P(Y|X)$

Examples of Discriminative Models

- Logistic regression
- Support vector machines(SVMs)
- Traditional neural networks
- Nearest neighbor

- Conditional Random Fields (CRFs)
- Decision Trees and Random Forest

1.6.2 The Approach of Generative Models

In the case of generative models, to find the conditional probability $P(Y|X)$, they estimate the prior probability $P(Y)$ and likelihood probability $P(X|Y)$ with the help of the training data and use the Bayes Theorem to calculate the posterior probability $P(Y|X)$:

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}} \Rightarrow P(Y|X) = \frac{P(Y) \cdot P(X|Y)}{P(X)}$$

Generative Models

Generative models are considered a class of statistical models that can generate new data instances. These models are used in unsupervised machine learning as a means to perform tasks such as

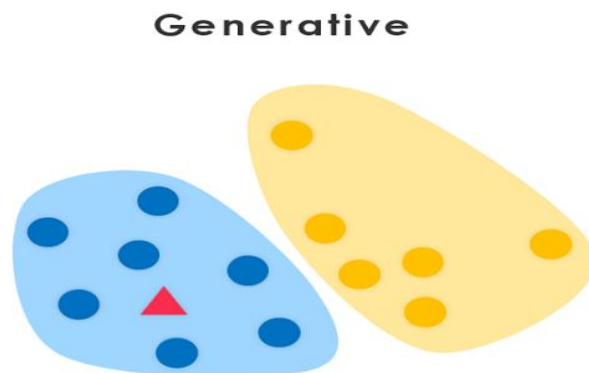
- Probability and Likelihood estimation,
- Modeling data points
- To describe the phenomenon in data,
- To distinguish between classes based on these probabilities.

Since these models often rely on the Bayes theorem to find the joint probability, generative models can tackle a more complex task than analogous discriminative models.

So, the Generative approach focuses on the distribution of individual classes in a dataset, and the learning algorithms tend to model the underlying patterns or distribution of the data points (e.g., gaussian). These models use the concept of joint probability and create instances where a given feature (x) or input and the desired output or label (y) exist simultaneously.

These models use probability estimates and likelihood to model data points and differentiate between different class labels present in a dataset. Unlike discriminative models, these models can also generate new data points.

However, they also have a major drawback – If there is a presence of outliers in the dataset, then it affects these types of models to a significant extent.



The Mathematics of Generative Models

Training generative classifiers involve estimating a function $f: X \rightarrow Y$, or probability $P(Y|X)$:

- Assume some functional form for the probabilities such as $P(Y)$, $P(X|Y)$
- With the help of training data, we estimate the parameters of $P(X|Y)$, $P(Y)$
- Use the Bayes theorem to calculate the posterior probability $P(Y|X)$

Examples of Generative Models

- Naïve Bayes
- Bayesian networks
- Markov random fields
- Hidden Markov Models (HMMs)
- Latent Dirichlet Allocation (LDA)
- Generative Adversarial Networks (GANs)
- Autoregressive Model

Difference between Discriminative and Generative Models

Let's see some of the differences between the Discriminative and Generative Models.

a. Core Idea

Discriminative models draw boundaries in the data space, while generative models try to model how data is placed throughout the space. A generative model explains how the data was generated, while a discriminative model focuses on predicting the labels of the data.

b. Mathematical Intuition

In mathematical terms, discriminative machine learning trains a model, which is done by learning parameters that maximize the conditional probability $P(Y|X)$. On the other hand, a generative model learns parameters by maximizing the joint probability of $P(X, Y)$.

c. Applications

Discriminative models recognize existing data, i.e., discriminative modeling identifies tags and sorts data and can be used to classify data, while Generative modeling produces something.

Since these models use different approaches to machine learning, both are suited for specific tasks i.e., Generative models are useful for unsupervised learning tasks. In contrast, discriminative models are useful for supervised learning tasks. GANs(Generative adversarial networks) can be thought of as a competition between the generator, which is a component of the generative model, and the discriminator, so basically, it is generative vs. discriminative model.

d. Outliers

Generative models have more impact on outliers than discriminative models.

e. Computational Cost

Discriminative models are computationally cheap as compared to generative models.

Comparison between Discriminative and Generative Models

Aspect	Generative Models	Discriminative Models
Purpose	Generate Data finding joint distribution $P(Y,X)$	Classify data finding decision boundary $P(Y X)$
Use Cases	Data generation, denoising, unsupervised learning	Classification, supervised learning tasks
Popular Models	Naïve Bayes, Bayesian networks, Markov random fields, Hidden Markov Models (HMMs)	Logistic Regression, Support Vector Machines
Training Focus	Maximize likelihood of observed data, Capture data structure	Learn decision boundary, Differentiate between classes
Example Task	Image generation, Inpainting (e.g., GANs, VAEs)	Text classification, Object detection (e.g., Deep Neural Networks)

Let's see some of the comparisons based on the following criteria between Discriminative and Generative Models:

a. Based on Performance

Generative models need fewer data to train compared with discriminative models since generative models are more biased as they make stronger assumptions, i.e., assumption of conditional independence.

b. Based on Missing Data

In general, if we have missing data in our dataset, then Generative models can work with these missing data, while discriminative models can't. This is because, in generative models, we can still estimate the posterior by marginalizing the unseen variables. However, discriminative models usually require all the features X to be observed.

c. Based on the Accuracy Score

If the assumption of conditional independence violates, then at that time, generative models are less accurate than discriminative models.

d. Based on Applications

Discriminative models are called “discriminative” since they are useful for discriminating Y 's label, i.e., target outcome, so they can only solve classification problems. In contrast, Generative models have more applications besides classification, such as samplings, Bayes learning, MAP inference, etc.

1.7 BASIC CONCEPTS IN MACHINE LEARNING

Terminologies used in machine learning:

- **Training example:** a sample from x including its output from the target function
- **Feature:** is a recorded property or a characteristic of examples. It is also referred to as “attribute”, or “variable” or “feature”
- **Target function:** the mapping function f from x to $f(x)$
- **Hypothesis:** approximation of f , a candidate function.
- **Concept:** A boolean target function, positive examples and negative examples for the 1/0 class values.
- **Classifier:** Learning program outputs a classifier that can be used to classify.
- **Learner:** Process that creates the classifier.
- **Hypothesis space:** set of possible approximations of f that the algorithm can create.
- **Version space:** subset of the hypothesis space that is consistent with the observed data.

1. Input representation

The general classification problem is concerned with assigning a class label to an unknown instance from instances of known assignments of labels. In a real world problem, a given situation or an object will have large number of features which may contribute to the assignment of the labels

2. Hypothesis

In a binary classification problem, a hypothesis is a statement or a proposition purporting to explain a given set of facts or observations.

3. Hypothesis space

The hypothesis space for a binary classification problem is a set of hypotheses for the problem that might possibly be returned by it.

4. Consistency and satisfying : Let x be an example in a binary classification problem and let $c(x)$ denote the class label assigned to x ($c(x)$ is 1 or 0). Let D be a set of training examples for the problem. Let h be a hypothesis for the problem and $h(x)$ be the class label assigned to x by the hypothesis h .

(a) We say that the hypothesis h is consistent with the set of training examples D if $h(x) = c(x)$ for all $x \in D$.

(b) We say that an example x satisfies the hypothesis h if $h(x) = 1$.

5. Ordering of hypotheses

Let X be the set of all possible examples for a binary classification problem and let h and h_1 be two hypotheses for the problem.

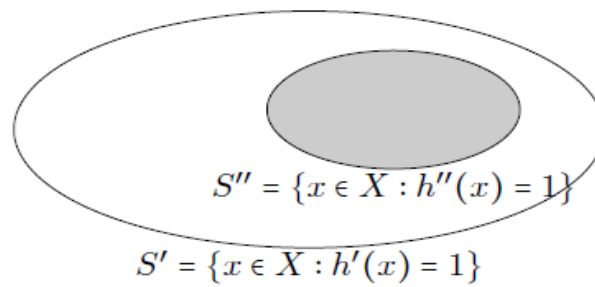


Figure : Hypothesis h^1 is more general than hypothesis h^{11} .. if and only if $S^{11} \subseteq S^1$

6. Version space

Consider a binary classification problem. Let D be a set of training examples and H a hypothesis space for the problem. The version space for the problem with respect to the set D and the space H is the set of hypotheses from H consistent with D ; that is, it is the set

$$VS_{D,H} = \{h \in H : h(x) = c(x) \text{ for all } x \in D\}.$$

7. Noise

Noise and their sources

Noise is any unwanted anomaly in the data . Noise may arise due to several factors:

1. There may be imprecision in recording the input attributes, which may shift the data points in the input space.
2. There may be errors in labeling the data points, which may relabel positive instances as negative and vice versa. This is sometimes called teacher noise.
3. There may be additional attributes, which we have not taken into account, that affect the label of an instance. Such attributes may be hidden or latent in that they may be unobservable. The effect of these neglected attributes is thus modeled as a random component and is included in “noise.”

8. Learning multiple classes

So far we have been discussing binary classification problems. In a general case there may be more than two classes. Two methods are generally used to handle such cases. These methods are known by the names “one-against-all” and “one-against-one”.

9. Model selection

In order to formulate a hypothesis for a problem, we have to choose some model and the term “model selection” has been used to refer to the process of choosing a model. However, the term has been used to indicate several things. In some contexts it has been used to indicate the process of choosing one particular approach from among several different approaches. This may be choosing an appropriate algorithms from a selection of possible algorithms, or choosing the sets of features to be used for input, or choosing initial values for certain parameters. Sometimes “model selection” refers

to the process of picking a particular mathematical model from among different mathematical models which all purport to describe the same data set. It has also been described as the process of choosing the right inductive bias.

10. Inductive bias

In a learning problem we only have the data. But data by itself is not sufficient to find the solution. We should make some extra assumptions to have a solution with the data we have. The set of assumptions we make to have learning possible is called the inductive bias of the learning algorithm. One way we introduce inductive bias is when we assume a hypothesis class.

11. Generalisation

How well a model trained on the training set predicts the right output for new instances is called generalization.

Generalization refers to how well the concepts learned by a machine learning model apply to specific examples not seen by the model when it was learning. The goal of a good machine learning model is to generalize well from the training data to any data from the problem domain. This allows us to make predictions in the future on data the model has never seen. **Overfitting and underfitting** are the two biggest causes for poor performance of machine learning algorithms. The model should be selected having the best generalisation. This is said to be the case if these problems are avoided.

- **Underfitting**

Underfitting is the production of a machine learning model that is not complex enough to accurately capture relationships between a dataset $\tilde{A}$'s features and a target variable.

- **Overfitting**

Overfitting is the production of an analysis which corresponds too closely or exactly to a particular set of data, and may therefore fail to fit additional data or predict future observations reliably.

1.8 THE MACHINE LEARNING PROCESS

It briefly examines the process by which machine learning algorithms can be selected, applied, and evaluated for the problem.

Steps in Machine Learning Process

1. Data Collection and Preparation
2. Feature Selection
3. Algorithm Choice
4. Parameter and Model Selection
5. Training
6. Evaluation

1. Data Collection and Preparation

Collect a reasonably small dataset with all of the features that you believe might be useful, and experiment with it before choosing the best features and collect and analyse the full dataset.

Often the difficulty is that there is a large amount of data that *might* be relevant, but it is hard to collect, either because it requires many measurements to be taken, or because they are in a variety of places and formats, and merging it appropriately is difficult, as is ensuring that it is clean; that is, it does not have significant errors, missing data, etc.

For supervised learning, target data is also needed, which can require the involvement of experts in the relevant field and significant investments of time.

Finally, the quantity of data needs to be considered. Machine learning algorithms need significant amounts of data, preferably without too much noise, but with increased dataset size comes increased computational costs, and the sweet spot at which there is enough data without excessive computational overhead is generally impossible to predict.

2. Feature Selection

An example of this part of the process was given in when we looked at possible features that might be useful for coin recognition. It consists of identifying the features that are most useful for the problem under examination. This invariably requires prior knowledge of the problem and the data; our common sense was used in the coins example above to identify some potentially useful features and to exclude others.

As well as the identification of features that are useful for the learner, it is also necessary that the features can be collected without significant expense or time, and that they are robust to noise and other corruption of the data that may arise in the collection process.

3. Algorithm Choice

Given the dataset, the choice of an appropriate algorithm (or algorithms) is what this book should be able to prepare you for, in that the knowledge of the underlying principles of each algorithm and examples of their use is precisely what is required for this.

4. Parameter and Model Selection

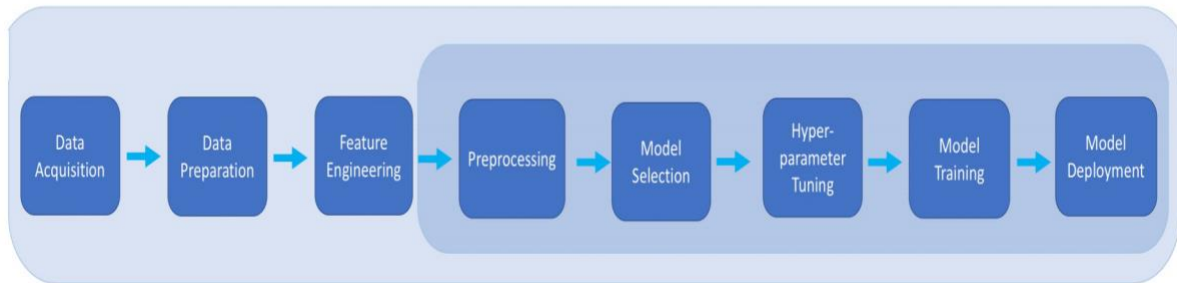
For many of the algorithms there are parameters that have to be set manually, or that require experimentation to identify appropriate values.

5. Training Given the dataset, algorithm, and parameters, training should be simply the use of computational resources in order to build a model of the data in order to predict the outputs on new data.

6. Evaluation

Before a system can be deployed it needs to be tested and evaluated for accuracy on data that it was not trained on. This can often include a comparison with human experts in the field, and the selection of appropriate metrics for this comparison.

Workflow of a (supervised) learning process



- **Data acquisition:** Get the training data, $D = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^m$
- **Exploratory Data Analysis (EDA):** Explore the data (e.g., by visualization, by computing basic statistics)
- **Data wrangling:** ... clean / scale / process the data / ...
- **Features:** Identify and process relevant features, define the feature space X
- **Inductive bias:** Choose a hypothesis class $\mathcal{H}$, define a hypothesis function f_{θ}
 - The hypothesis function f_{θ} is the model of the ML task, it is defined by a vector of parameters $\theta \in \mathbb{R}^n$
- **Loss function for model evaluation:** Select an appropriate loss function, ℓ , to evaluate the quality of f_{θ}
- **Train the model:** Use the data in D to iteratively set the values of the parameter vector θ such that the loss produced by f_{θ} on the training set (?) is minimized
- ...

3

1.8 REINFORCEMENT LEARNING

Reinforcement learning

- Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance.
- Agent gets rewarded for each good action and gets punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards.
- In reinforcement learning, there is no labelled data like supervised learning, and agents learn from their experiences only.

- The reinforcement learning process is similar to a human being; for example, a child learns various things by experiences in his day-to-day life.
- An example of reinforcement learning is to play a game, where the Game is the environment, moves of an agent at each step define states, and the goal of the agent is to get a high score.
- Agent receives feedback in terms of punishment and rewards. Due to its way of working, reinforcement learning is employed in different fields such as Game theory, Operation Research, Information theory, multi-agent systems.
- A reinforcement learning problem can be formalized using Markov Decision Process(MDP). In MDP, the agent constantly interacts with the environment and performs actions; at each action, the environment responds and generates a new state.

Figure shows how they are linked with each other and with the reward.

In reinforcement learning the algorithm gets feedback in the form of the reward about how well it is doing. In contrast to supervised learning, where the algorithm is ‘taught’ the correct answer, the reward function evaluates the current solution, but does not suggest how to improve it. We therefore need to allow for rewards that don’t appear until long after the relevant actions have been taken. Sometimes we think of the immediate reward and the total expected reward into the future.

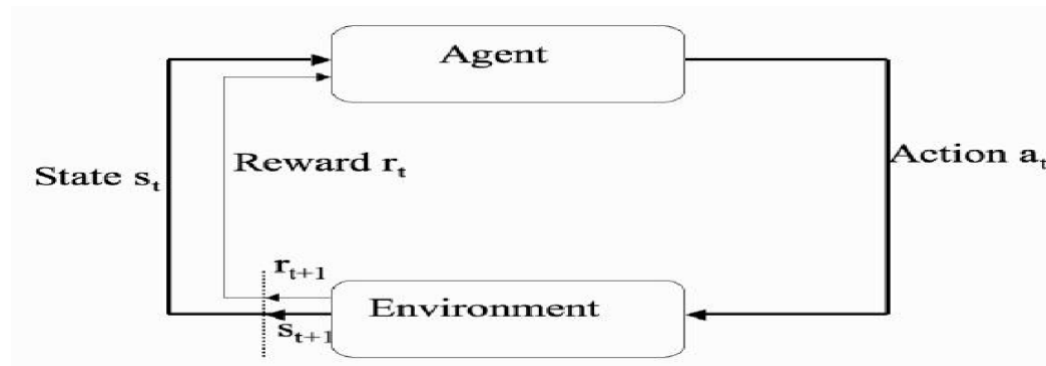


FIGURE The reinforcement learning cycle: the learning agent performs action a_t in state s_t and receives reward r_{t+1} from the environment, ending up in state s_{t+1} .

Once the algorithm has decided on the reward, it needs to choose the action that should be performed in the current state. This is known as the policy. This is done based on some combination of exploration and exploitation (remember, reinforcement learning is basically a search method), which in this case means deciding whether to take the action that gave the highest reward last time we were in this state, or trying out a different action in the hope of finding something even better.

Advantages of Reinforcement Learning: -

- It helps in solving complex real-world problems which are difficult to be solved by general techniques.
- The learning model of RL is similar to the learning of human beings; hence most accurate results can be found.

- Helps in achieving long term results.

Disadvantage of Reinforcement Learning: -

- RL algorithms are not preferred for simple problems.
- RL algorithms require huge data and computations.
- Too much reinforcement learning can lead to an overload of states which can weaken the results.
- The curse of dimensionality limits reinforcement learning for real physical systems.

Some of the important reinforcement learning algorithms are:

1. Q-learning
2. Sarsa
3. Monte Carlo
4. Deep Q network

Applications

Reinforcement learning algorithms are widely used in the gaming industries to build games. It is also used to train robots to do human tasks.



This is where the name ‘reinforcement learning’ comes from, since you repeat actions that are reinforced by a feeling of satisfaction.